

Unauthorized RDP Access Leading to Domain Controller Compromise and Possible Data Exfiltration

DATE OF REPORT

2025-10-11

INCIDENT DATE

2025-10-07

REPORT STATUS

Escalated

REPORTED BY

Salman Rathor

REVIEWED BY

Steven · DFIR Lead

REVIEW DATE

2025-10-17

SEVERITY: CRITICAL

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1 Report Title

Unauthorized RDP Access Leading to Domain Controller Compromise and Possible Data Exfiltration

2 Date of Report

2025-10-11

3 Reported By

Salman Rathor

4 Severity Level

Critical

Severity justification: An unauthorized user obtained administrative access to a workstation, disabled security controls, established multiple persistence mechanisms, introduced Mimikatz, accessed the domain controller, extracted Active Directory credential information, collected sensitive files and possibly exfiltrated information.

5 Summary of Findings

On October 7, 2025, an unauthorized individual gained access to MTS-CONTRACTORPC1 through RDP using the Administrator account. After gaining access, the attacker used PowerShell to download suspicious scripts and executables from external infrastructure.

The attacker disabled Windows Defender, Windows Firewall and Windows event logging before establishing multiple persistence mechanisms through registry Run keys, scheduled tasks and a permanent WMI event subscription. Mimikatz was subsequently downloaded and staged for credential-access activity.

The attacker later accessed the MTS domain controller, imported the Active Directory PowerShell module and executed multiple suspicious programs. Sensitive files were accessed, a Volume Shadow Copy was created, and Active Directory database information was extracted using `ntdsutil.exe`.

The collected information was archived in `C:\ProgramData\Microsoft\Crypto\RSA\backup.zip`. Activity involving `file.io` and a DNS query containing Base64-encoded KRBTGT-related information indicated possible data exfiltration.

The attacker then cleared Windows event logs, deleted shadow copies and removed PowerShell command-history information. Based on the compromise of the domain controller and extraction of Active Directory credential information, the environment should be treated as experiencing a full domain compromise until containment and recovery are completed.

6 Investigation Timeline

Timezone: UTC

Date and time	Event description
2025-10-07 04:13:10	External IP address 81.141.200.165 successfully authenticated to MTS-CONTRACTORPC1 through RDP using the Administrator account.
2025-10-07 04:18:54	PowerShell established an outbound connection to 78.141.234.86 and downloaded kb5029244.ps1.
2025-10-07 04:19:01	Windows Firewall was disabled across all profiles using netsh.exe advfirewall set allprofiles state off.
2025-10-07 04:19:01-06:21:46	Windows Defender exclusions were created for eight processes and several directories, including C:\Windows\Temp, C:\Users\Public, C:\ProgramData and C:\Finance.
2025-10-07 06:00:20-06:21:46	Four registry values were modified to disable Windows Defender features.
2025-10-07 06:01:13	A second attempt was made to disable Windows Firewall.
2025-10-07 06:01:15	System discovery commands, including whoami, were executed.
2025-10-07 06:16:36	Registry Run-key persistence was created using the value name OneDriveStandAlone.
2025-10-07 06:16:38	Scheduled tasks named GoogleUpdateTaskMachineCore and MicrosoftEdgeUpdateTaskMachineUA were created.
2025-10-07 06:16:38	A WMI Filter-Consumer binding and WMI consumer named WindowsSystemHealthConsumer were created for persistence.
2025-10-07 06:21:01	PowerShell launched C:\Users\Public\MicrosoftEdgeUpdate.exe using a hidden window.
2025-10-07 06:21:46	regsvr32.exe was added to the Windows Defender exclusion list.
2025-10-07 06:21:47	Another attempt was made to disable Windows Firewall.
2025-10-07 06:21:49	Windows event logging was disabled on MTS-CONTRACTORPC1.
2025-10-07 06:27:22	Mimikatz was downloaded.

Date and time	Event description
2025-10-07 06:27:24	Mimikatz and update.bin were staged in C:\Windows\Temp\TMP121235\.
2025-10-07 06:32:26	MTS-CONTRACTORPC1 connected to 78.141.234.86 over destination port 1337 and downloaded mimikatz.exe.
2025-10-07 06:42:44	Additional scheduled tasks named SystemHealthMonitor and QuickBooksSync were created to execute suspicious payloads.
2025-10-07 07:02:33	The Administrator account was used to access the MTS domain controller. The source was recorded as public IP address 74.249.82.128.
2025-10-07 07:04:29	PowerShell imported the Active Directory module using Import-Module ActiveDirectory.
2025-10-07 07:07:00	The same PowerShell session launched svchost_update.exe, MicrosoftEdgeUpdate.exe and quickbooks_sync.exe.
2025-10-07 07:17:54–08:00:35	Multiple sensitive files were accessed, indicating data collection.
2025-10-07 07:35:52	Mimikatz-related file activity was identified on the domain controller.
2025-10-07 08:05:43	A Volume Shadow Copy of the domain controller's C: drive was created using vssadmin.exe.
2025-10-07 08:05:46	Active Directory database information was extracted using ntdsutil.exe.
2025-10-07 08:10:59	Collected information was archived into C:\ProgramData\Microsoft\Crypto\RSA\backup.zip.
2025-10-07 08:11:01	Activity indicated a possible upload of collected information to file.io.
2025-10-07 08:11:01	nslookup.exe sent a DNS query containing Base64-encoded KRBTGT-related information through Google DNS.
2025-10-07 08:16:01	Windows Security, Application and System event logs were cleared.
2025-10-07 08:16:05	Shadow copies were deleted using vssadmin.exe.
2025-10-07 08:16:05	The PowerShell command-history file was deleted.

7 Who, What, When, Where, Why and How

Who

The attacker used the Administrator account.

The confirmed affected systems were:

- MTS-CONTRACTORPC1 — 192.168.10.11
- MTS-DC.mts.local — 192.168.10.8

External infrastructure associated with the activity included:

- 81.141.200.165 — Source of the initial RDP authentication
- 78.141.234.86 — Source of suspicious scripts and Mimikatz
- 74.249.82.128 — Recorded source of access to the domain controller
- 20.246.83.211 — Public IP address associated with the domain controller
- file.io — Possible data-exfiltration destination
- Google DNS — Used for a DNS query containing encoded information

The real identity of the attacker was not determined.

What

An attacker gained unauthorized remote access to ContractorPC1 using valid Administrator credentials. The attacker downloaded suspicious payloads, disabled security controls, established persistence, introduced Mimikatz, accessed the domain controller, extracted Active Directory credential information, collected sensitive files and possibly exfiltrated data.

The attacker also deleted logs and command history in an apparent attempt to conceal the activity.

When

The first confirmed suspicious activity occurred on October 7, 2025, at 4:13:10 AM.

The final observed activity occurred on October 7, 2025, at approximately 8:16:05 AM.

The confirmed incident duration was approximately four hours and three minutes.

The timezone must be confirmed from the original log sources.

Where

The initial compromise occurred on:

- MTS-CONTRACTORPC1
- Internal IP: 192.168.10.11

The attacker subsequently accessed:

- MTS-DC.mts.local
- Internal IP: 192.168.10.8

Suspicious files and collected data were stored in locations including:

- C:\Users\Public
- C:\Windows\Temp
- C:\Windows\Temp\TMP121235
- C:\ProgramData\Microsoft\Crypto\RSA

Why

The attacker gained access by successfully authenticating through RDP using the Administrator account. The available evidence does not establish how the attacker originally obtained the Administrator credentials.

The attacker's likely objectives were to:

- Maintain persistent access
- Disable security protections
- Obtain additional credentials
- Compromise Active Directory
- Collect sensitive files
- Transfer collected information outside the environment
- Remove evidence of the intrusion

How

The attacker used RDP for initial access and PowerShell to download and execute suspicious payloads.

Windows Defender, Windows Firewall and event logging were disabled to reduce detection. Registry Run keys, scheduled tasks and WMI event subscriptions were used to establish persistence.

Mimikatz was introduced for credential-access activity. The attacker then accessed the domain controller, imported the Active Directory PowerShell module and executed suspicious binaries.

The attacker created a shadow copy and used `ntdsutil.exe` to extract Active Directory database information. Sensitive files and credential-related data were placed into a ZIP archive and possibly transferred through `file.io` and DNS queries.

Finally, the attacker cleared Windows event logs, deleted shadow copies and removed PowerShell command history.

8 MITRE ATT&CK; Techniques

Tactic	Technique	Technique ID	Evidence
Initial Access	External Remote Services	T1133	External RDP authentication to ContractorPC1
Execution	PowerShell	T1059.001	PowerShell downloaded and launched suspicious payloads
Command and Control	Ingress Tool Transfer	T1105	Scripts and Mimikatz were downloaded from external infrastructure
Defense Evasion	Impair Defenses	T1562.001	Windows Defender settings and exclusions were modified
Defense Evasion	Disable or Modify System Firewall	T1562.004	Windows Firewall was disabled across all profiles
Defense Evasion	Disable Windows Event Logging	T1562.002	Event logging was disabled on ContractorPC1
Discovery	System Owner/User Discovery	T1033	whoami was executed

Tactic	Technique	Technique ID	Evidence
Persistence	Registry Run Keys / Startup Folder	T1547.001	OneDriveStandAlone Run key was created
Persistence	Scheduled Task/Job: Scheduled Task	T1053.005	Multiple scheduled tasks were created
Persistence	Event Triggered Execution: WMI Event Subscription	T1546.003	WMI Filter-Consumer binding was created
Credential Access	OS Credential Dumping	T1003	Mimikatz was downloaded and staged
Credential Access	OS Credential Dumping: NTDS	T1003.003	Active Directory information was extracted using ntdsutil.exe
Lateral Movement	Remote Services: RDP	T1021.001	Administrator account accessed the domain controller
Discovery	Account Discovery: Domain Account	T1087.002	Active Directory PowerShell module was imported
Collection	Data from Local System	T1005	Sensitive files were accessed
Collection	Archive Collected Data	T1560	Collected files were placed into backup.zip
Exfiltration	Exfiltration Over Web Service	T1567	Possible upload to file.io
Exfiltration	Exfiltration Over Alternative Protocol	T1048	Encoded KRBTGT information was sent through DNS
Defense Evasion	Clear Windows Event Logs	T1070.001	Security, Application and System logs were cleared
Defense Evasion	Clear Command History	T1070.003	PowerShell history was deleted
Impact	Inhibit System Recovery	T1490	Shadow copies were deleted using vssadmin.exe

9 Impact Assessment

Systems affected

The following systems were confirmed affected:

- MTS-CONTRACTORPC1
- MTS-DC.mts.local

The compromise expanded from a contractor workstation to the organization's domain controller.

Accounts affected

The Administrator account was confirmed to have been used by the attacker.

Because Mimikatz and Active Directory database extraction were observed, credentials associated with accounts that authenticated to the affected systems should be considered potentially compromised.

Domain password hashes, Kerberos information and privileged authentication material may have been exposed.

Data affected

Sensitive files were accessed and placed into an archive named `backup.zip`.

Active Directory credential information was extracted using a shadow copy and `ntdsutil.exe`.

Possible data exfiltration was identified through:

- Communication with `file.io`
- A DNS query containing Base64-encoded KRBTGT-related information

The available evidence does not conclusively confirm the complete contents or successful transfer of all collected data.

Security impact

The compromise of the domain controller created a risk that the attacker could:

- Obtain domain password hashes
- Impersonate domain users
- Access additional systems
- Create or modify privileged accounts
- Maintain persistence within the domain
- Conduct additional lateral movement
- Regain access after basic password resets
- Access confidential organizational data

Business impact

Potential business consequences include:

- Exposure of confidential information
- Unauthorized access to internal systems
- Account compromise across the domain
- Operational disruption during containment and rebuilding
- Regulatory or contractual reporting obligations
- Reputational damage
- Increased recovery and forensic-investigation costs

Overall impact conclusion

The incident represents a critical compromise of the Active Directory environment. The organization should treat all privileged credentials and authentication material present on the affected systems as potentially compromised until a complete domain recovery process is performed.

10 Recommendations and Next Steps

Immediate containment

Priority	Recommendation	Suggested owner
IMMEDIATE	Isolate ContractorPC1 from the network while preserving forensic evidence.	Incident Response / IT
IMMEDIATE	Isolate or restrict the domain controller while maintaining required business services.	Active Directory / Incident Response
IMMEDIATE	Disable the compromised Administrator account.	Identity Team
IMMEDIATE	Revoke active RDP, VPN, cloud and domain sessions associated with affected accounts.	Identity Team
IMMEDIATE	Block 81.141.200.165 and 78.141.234.86 at applicable security controls.	Network Security
IMMEDIATE	Block confirmed malicious file hashes, domains and URLs.	Endpoint / Network Security
IMMEDIATE	Preserve endpoint, firewall, DNS, proxy, Active Directory and cloud logs.	SOC / DFIR

Credential and domain recovery

Priority	Recommendation	Suggested owner
IMMEDIATE	Reset the compromised Administrator password.	Identity Team
IMMEDIATE	Reset all privileged accounts used on the affected systems.	Identity Team
HIGH	Reset service-account credentials that may have been exposed.	Application / Identity Teams
HIGH	Perform a controlled two-stage KRBTGT password reset.	Active Directory Team
HIGH	Review privileged-group membership and recently created or modified accounts.	Active Directory Team
HIGH	Investigate suspicious Kerberos ticket activity and replication-related activity.	SOC / DFIR

Eradication and recovery

Priority	Recommendation	Suggested owner
HIGH	Remove the OneDriveStandAlone Run-key persistence mechanism.	Incident Response
HIGH	Remove all malicious scheduled tasks.	Incident Response
HIGH	Remove the malicious WMI filter, consumer and binding.	Incident Response
HIGH	Remove suspicious files and tools, including Mimikatz and associated payloads.	Incident Response
HIGH	Restore Windows Defender, Windows Firewall and event-logging settings.	Endpoint Security
HIGH	Remove unauthorized Windows Defender exclusions.	Endpoint Security

Priority	Recommendation	Suggested owner
HIGH	Rebuild affected systems from known-good images if integrity cannot be guaranteed.	Infrastructure Team
HIGH	Confirm that all persistence mechanisms were removed before reconnecting systems.	Incident Response

Exfiltration investigation

Priority	Recommendation	Suggested owner
HIGH	Review proxy and firewall logs for connections to file.io.	SOC / Network Security
HIGH	Review DNS logs for additional encoded or unusually long DNS queries.	SOC / DNS Team
HIGH	Determine whether backup.zip was successfully transferred outside the environment.	DFIR
HIGH	Identify the exact sensitive files accessed between 7:17:54 AM and 8:00:35 AM.	DFIR / Data Owners
HIGH	Determine whether notification or regulatory reporting obligations apply.	Legal / Privacy / Management

Long-term security improvements

Priority	Recommendation	Suggested owner
MEDIUM	Remove direct public exposure of RDP.	Network Security
MEDIUM	Require MFA for all remote and privileged access.	Identity Team
MEDIUM	Restrict access to domain controllers to approved administrative systems.	Active Directory / Network Teams
MEDIUM	Separate normal user accounts from privileged administrative accounts.	Identity Team
MEDIUM	Enable Windows Defender tamper protection.	Endpoint Security
MEDIUM	Alert on Defender exclusions, firewall disablement and event-log disablement.	SOC / Endpoint Security
MEDIUM	Alert on Mimikatz, LSASS access, vssadmin.exe and ntdsutil.exe.	SOC
MEDIUM	Monitor the creation of Run keys, scheduled tasks and WMI subscriptions.	SOC
MEDIUM	Restrict outbound internet access from domain controllers.	Network Security
MEDIUM	Block unauthorized file-sharing services.	Network Security
MEDIUM	Monitor DNS traffic for encoded or unusually long queries.	SOC / DNS Team
MEDIUM	Improve centralized and tamper-resistant log collection.	Security Engineering

11 Attachments and Evidence

Evidence ID	Description
E-01	RDP authentication event showing successful Administrator login from 81.141.200.165
E-02	PowerShell network event showing connection to 78.141.234.86
E-03	Download event for kb5029244.ps1
E-04	Windows Defender exclusion and configuration-change events
E-05	Windows Firewall disablement events
E-06	Registry Run-key persistence event for OneDriveStandAlone
E-07	Scheduled-task creation events
E-08	WMI Filter-Consumer binding and consumer details
E-09	Execution event for MicrosoftEdgeUpdate.exe
E-10	Mimikatz download and file-creation events
E-11	Domain-controller authentication event
E-12	Active Directory PowerShell module execution
E-13	Execution events for suspicious files on the domain controller
E-14	Sensitive-file access records
E-15	Shadow-copy creation using vssadmin.exe
E-16	Active Directory extraction using ntdsutil.exe
E-17	Creation of backup.zip
E-18	Network activity involving file.io
E-19	DNS query containing encoded KRBGT-related information
E-20	Windows event-log clearing records
E-21	Shadow-copy deletion activity
E-22	PowerShell history-file deletion event

12 Report Status

ESCALATED

The incident should remain escalated to the DFIR, Incident Response and Active Directory teams until the following are confirmed:

- Affected systems have been contained

- Persistence has been removed
- Privileged credentials have been rotated
- The KRBTGT recovery procedure has been completed
- Data-exfiltration scope has been determined
- Affected systems have been safely restored
- No additional compromised systems or accounts remain

The report may be changed to Closed after containment, eradication, recovery and final management review are completed.

13 Reviewed By

Reviewer name	Reviewer title	Review date
Steven	DFIR Lead	2025-10-17

Appendix A — Evidence Screenshots

The exhibits below correspond to the evidence identifiers listed in Section 11. Screenshots are from the original investigation workspace.

E-01 — Successful RDP Authentication

TimeGenerated	Account	EventID	IpAddress	LogonType
> Oct 7, 2025 4:13:08 AM	MTS-CONTRACTORP\ad...	4624	(info) 81.141.200.165	3
> Oct 7, 2025 4:13:10 AM	MTS-CONTRACTORP\ad...	4624	(info) 81.141.200.165	10
> Oct 7, 2025 4:13:10 AM	MTS-CONTRACTORP\ad...	4624	(info) 81.141.200.165	10

Figure E-01: Authentication logs showing external IP address 81.141.200.165 successfully logging into ContractorPC1 through RDP using the Administrator account at 4:13:10 AM.

E-02 — PowerShell External Connection

16
17 DeviceNetworkEvents_CL
18 |where Lab == "CTF-1"
19 |where Timestamp between (datetime(2025-10-07 00:00) .. datetime(2025-10-08 00:00))
20 |where DeviceName == "mts-contractorpc1"
21 |where InitiatingProcessFileName contains "powershell"
22 |project Timestamp, LocalIP, RemoteIP, InitiatingProcessFileName, RemoteUrl
23

Results Query history Getting started

Export Show empty columns 2 items Search

Filters: Add filter

	Timestamp	LocalIP	RemoteIP	InitiatingProcessFil...	RemoteUrl
☐	> Oct 7, 2025 4:18:5...	192.168.10.11	78.141.234.86	powershell.exe	
☐	> Oct 7, 2025 4:18:5...	192.168.10.11	78.141.234.86	powershell.exe	http://78.141.234.86:1337/kb5029244.ps1

Timestamp Oct 7, 2025 4:18:54 AM
LocalIP 192.168.10.11

Figure E-02: Network telemetry showing PowerShell connecting to 78.141.234.86

E-03 — PowerShell Script Downloaded

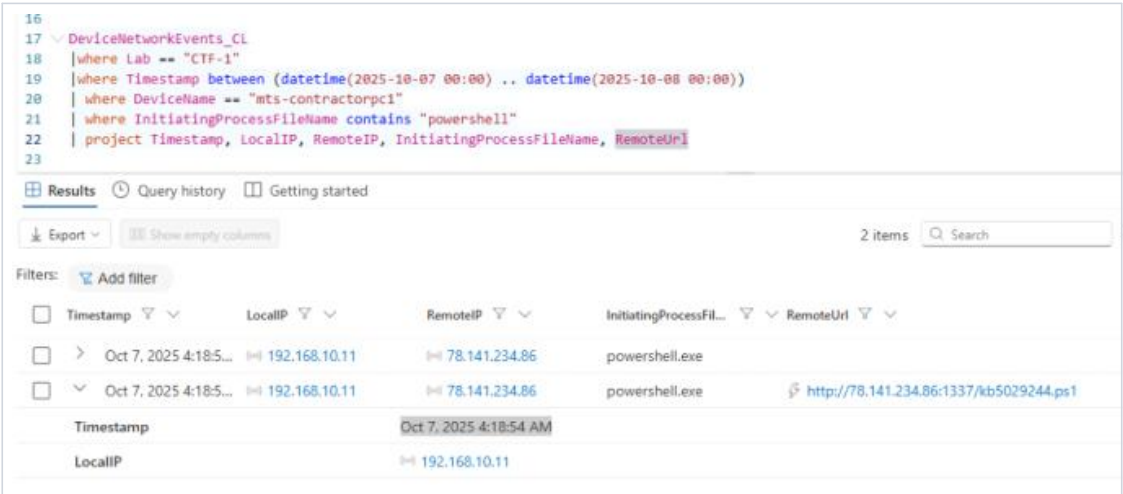
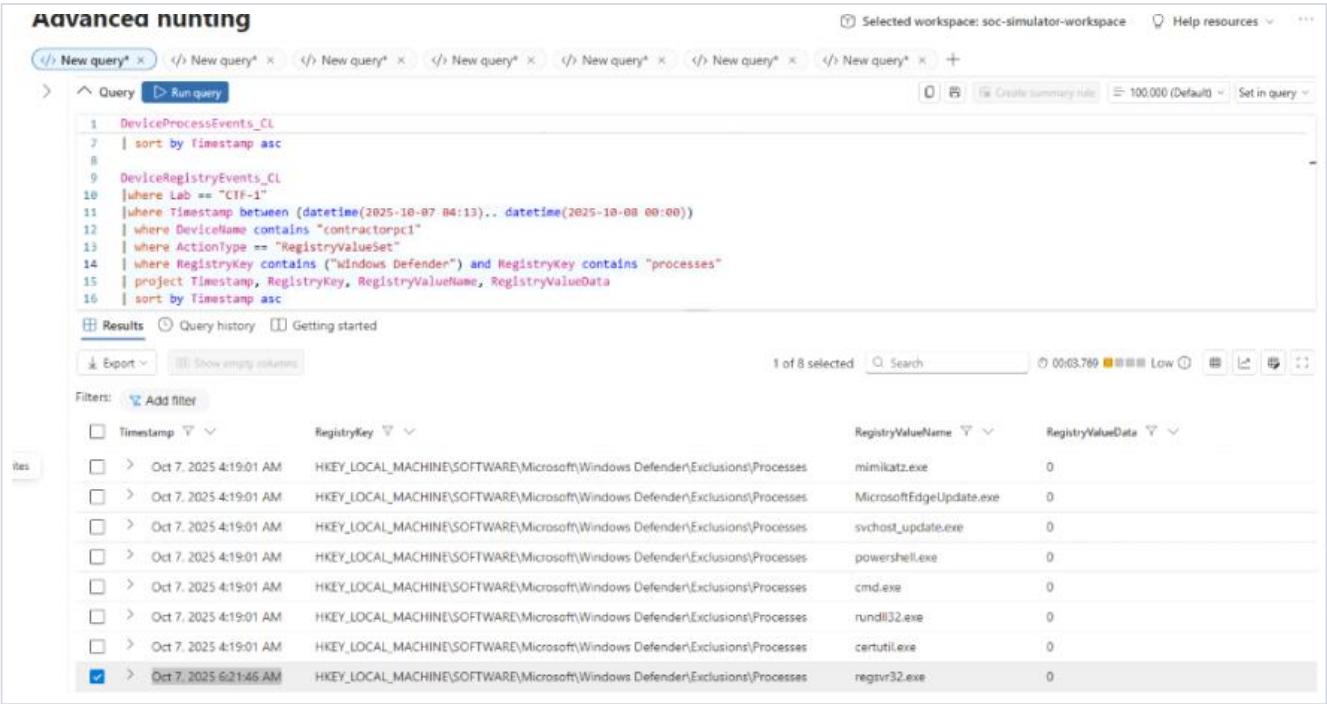


Figure E-03: File or process telemetry showing the download of kb5029244.ps1.

E-04 — Windows Defender Configuration Changes



☐	Timestamp ▾ ▾	RegistryKey ▾ ▾	RegistryValueName ▾ ▾	RegistryValueData ▾ ▾
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection	DisableIOAVProtection	1
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\Windows\Temp	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\Users\Public	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\ProgramData	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\Finance	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes	cmd.exe	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes	mimikatz.exe	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Spynet	SpyNetReporting	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection	DisableRealtimeMonitoring	1
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes	certutil.exe	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes	svchost_update.exe	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\Users\ADMINI~1\AppData\Local\Temp	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection	DisableBehaviorMonitoring	1
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\Users\administrator\AppData\Roaming	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes	rundll32.exe	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\CTF_Data	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\Users\administrator\AppData\Local	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes	MicrosoftEdgeUpdate.exe	0
☐	> Oct 7, 2025 4:19:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes	powershell.exe	0
☐	> Oct 7, 2025 6:00:1...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Extensions	.ps1	0
☐	> Oct 7, 2025 6:00:1...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths	C:\Users	0
☐	> Oct 7, 2025 6:00:1...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Extensions	.exe	0

>	Oct 7, 2025 6:00:2...	HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender	DisableAntiSpyware	1
>	Oct 7, 2025 6:21:4...	HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender\Real-Time Pro...	DisableIOAVProtection	1
>	Oct 7, 2025 6:21:4...	HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender	DisableAntiVirus	1
>	Oct 7, 2025 6:21:4...	HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender\Real-Time Pro...	DisableBehaviorMonitoring	1
>	Oct 7, 2025 6:21:4...	HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender\Exclusions\Processes	regsvr32.exe	0
>	Oct 7, 2025 7:10:0...	HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows Defender\Real-Time Pro...	DisableBehaviorMonitoring	0

Figures E-04: Windows Defender configuration-change events showing the creation of process and directory exclusions and the disabling of multiple antivirus protection features on MTS-CONTRACTORPC1.

E-05 — Windows Firewall Disabled

Screenshot not included in the source report.

E-06 — Registry Run-Key Persistence

9	DeviceRegistryEvents_CL
11	where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
12	where DeviceName contains "contractorpc1"
13	where ActionType == "RegistryValueSet"
14	where RegistryKey has_any (@"\Run", @"\RunOnce")
15	project Timestamp, RegistryKey, RegistryValueName, RegistryValueData
16	sort by Timestamp asc
17	

Results

Query history

Getting started

Export

Show empty columns

1 of 10 selected

Search

00:01.612

Low

Filters:

Add filter

	Timestamp	RegistryKey	RegistryValueName	RegistryValueData
☐	> Oct 7, 2025 4:13:12 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	ctfmon.exe	ctfmon.exe /n
☐	> Oct 7, 2025 4:18:07 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	Delete Cached Update B...	C:\Windows\system32\cmd.exe /q /c del /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\Updat...
☐	> Oct 7, 2025 4:18:07 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	Delete Cached Standalo...	C:\Windows\system32\cmd.exe /q /c del /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\Standa...
☐	> Oct 7, 2025 4:18:16 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	Uninstall 21.220.1024.00...	C:\Windows\system32\cmd.exe /q /c rmdir /s /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\2'
☐	> Oct 7, 2025 4:18:16 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	Uninstall 21.220.1024.00...	C:\Windows\system32\cmd.exe /q /c rmdir /s /q "C:\Users\administrator\AppData\Local\Microsoft\OneDrive\2'
☐	> Oct 7, 2025 6:16:36 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	MicrosoftEdgeUpdateC...	"C:\Users\Public\MicrosoftEdgeUpdate.exe"
☒	> Oct 7, 2025 6:16:36 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	OneDriveStandalone	powershell.exe -NoP -W Hidden -c Start-Process 'C:\Users\Public\MicrosoftEdgeUpdate.exe'
☐	> Oct 7, 2025 6:42:44 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	QuickBooksAutoSync	C:\Users\administrator\AppData\Roaming\QuickBooksSync\quickbooks_sync.exe
☐	> Oct 7, 2025 6:42:44 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	WindowsDefenderUpdate	C:\Users\Public\SageTaxTools\svchost_update.exe
☐	> Oct 7, 2025 6:42:44 AM	HKEY_CURRENT_USER\S-1-5-21-2718...	MicrosoftEdgeUpdateC...	C:\Users\Public\MicrosoftEdgeUpdate.exe

Figure E-06: Registry telemetry showing the creation of a Run-key persistence entry named OneDriveStandalone on MTS-CONTRACTORPC1.

E-07 — Scheduled Task Persistence

21	DeviceEvents_CL
22	where Lab == "CTF-1"
23	where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
24	where DeviceName contains "contractorpc1"
25	where ActionType == "ScheduledTaskCreated"
26	/// where ActionType contains "ReadProcessMemoryApiCall"
27	/// where FileName contains "lsass"
28	/// project Timestamp, InitiatingProcessFileName, FileName, InitiatingProcessCommandLine
29	project Timestamp, AdditionalData, InitiatingProcessFileName, FileName, ActionType
30	sort by Timestamp asc

Results

Query history

Getting started

Export

Show empty columns

6 items

Search

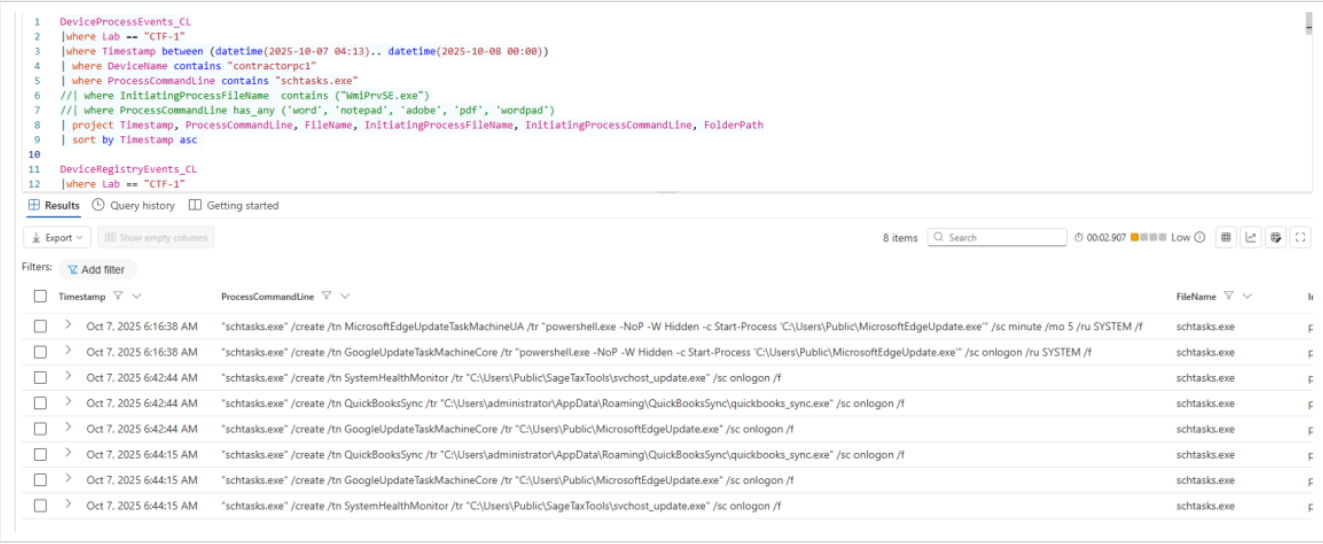
00:01.483

Low

Filters:

Add filter

	Timestamp	AdditionalData	Initia
☐	> Oct 7, 2025 4:18:07 ...	("TaskName": "\\OneDrive Standalone Update Task-S-1-5-21-2718350712-1533534050-804694281-1000", "TaskContent": {"Triggers": {"TimeTrigger": {"StartBoundary": "1992-05-01T04:00:00", "Enabled": true, "Rep...	
☐	> Oct 7, 2025 4:18:07 ...	("TaskName": "\\OneDrive Startup Task-S-1-5-21-2718350712-1533534050-804694281-1000", "TaskContent": {"Triggers": {"LogonTrigger": {"ExecutionTimeLimit": "PT5M", "Enabled": true, "UserId": "S-1-5-21-2718...	
☐	> Oct 7, 2025 6:16:38 ...	("TaskName": "\\GoogleUpdateTaskMachineCore", "TaskContent": {"Triggers": {"LogonTrigger": {"StartBoundary": "2025-10-07T06:16:00", "Enabled": true, "Actions": {"Exec": {"Command": "powershell.exe", "Ar...	
☐	> Oct 7, 2025 6:16:38 ...	("TaskName": "\\MicrosoftEdgeUpdateTaskMachineUA", "TaskContent": {"Triggers": {"TimeTrigger": {"Repetition": {"Interval": "PT5M", "StopAtDurationEnd": false, "StartBoundary": "2025-10-07T06:16:00", "Enabl...	
☐	> Oct 7, 2025 6:42:44 ...	("TaskName": "\\QuickBooksSync", "TaskContent": {"Triggers": {"LogonTrigger": {"StartBoundary": "2025-10-07T06:42:00", "Enabled": true, "Actions": {"Exec": {"Command": "C:\\Users\\administrator\\AppD...	
☐	> Oct 7, 2025 6:42:44 ...	("TaskName": "\\SystemHealthMonitor", "TaskContent": {"Triggers": {"LogonTrigger": {"StartBoundary": "2025-10-07T06:42:00", "Enabled": true, "Actions": {"Exec": {"Command": "C:\\Users\\Public\\SageTax...	



Figures E-07: Scheduled-task creation events showing the creation of suspicious tasks, including GoogleUpdateTaskMachineCore, MicrosoftEdgeUpdateTaskMachineUA, SystemHealthMonitor and QuickBooksSync.

E-08 — WMI Event-Subscription Persistence

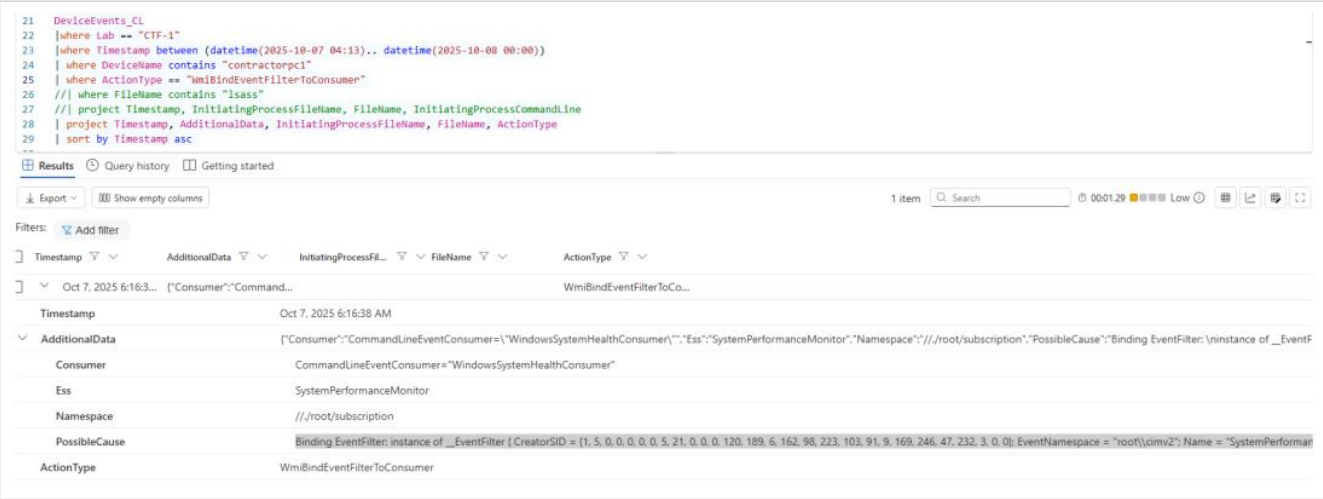


Figure E-08: WMI telemetry showing the creation of a malicious event filter, command-line consumer and Filter-to-Consumer binding associated with WindowsSystemHealthConsumer.

E-09 — Suspicious Executable Launched

Screenshot not included in the source report.

E-10 — Mimikatz Downloading and Staging

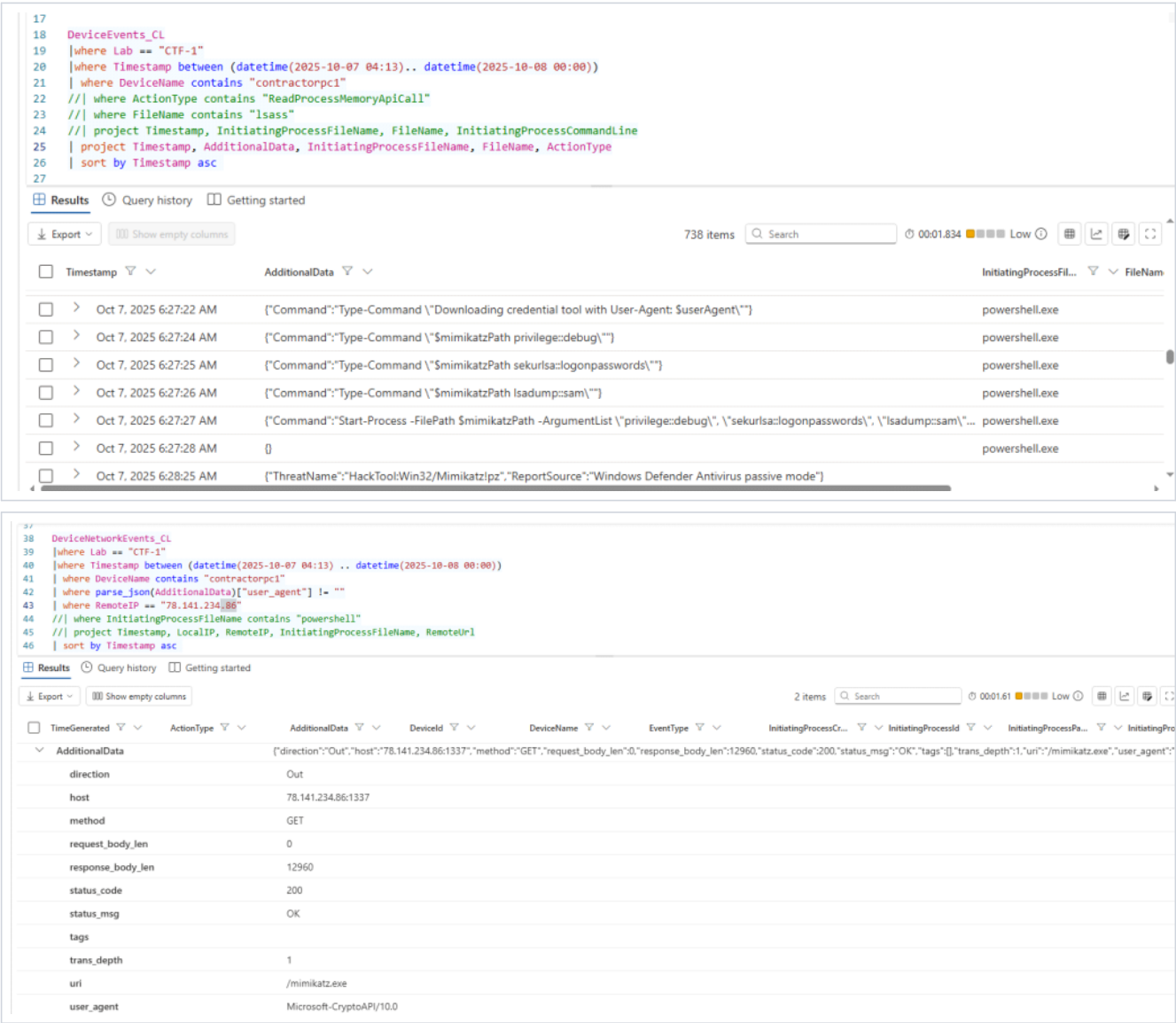
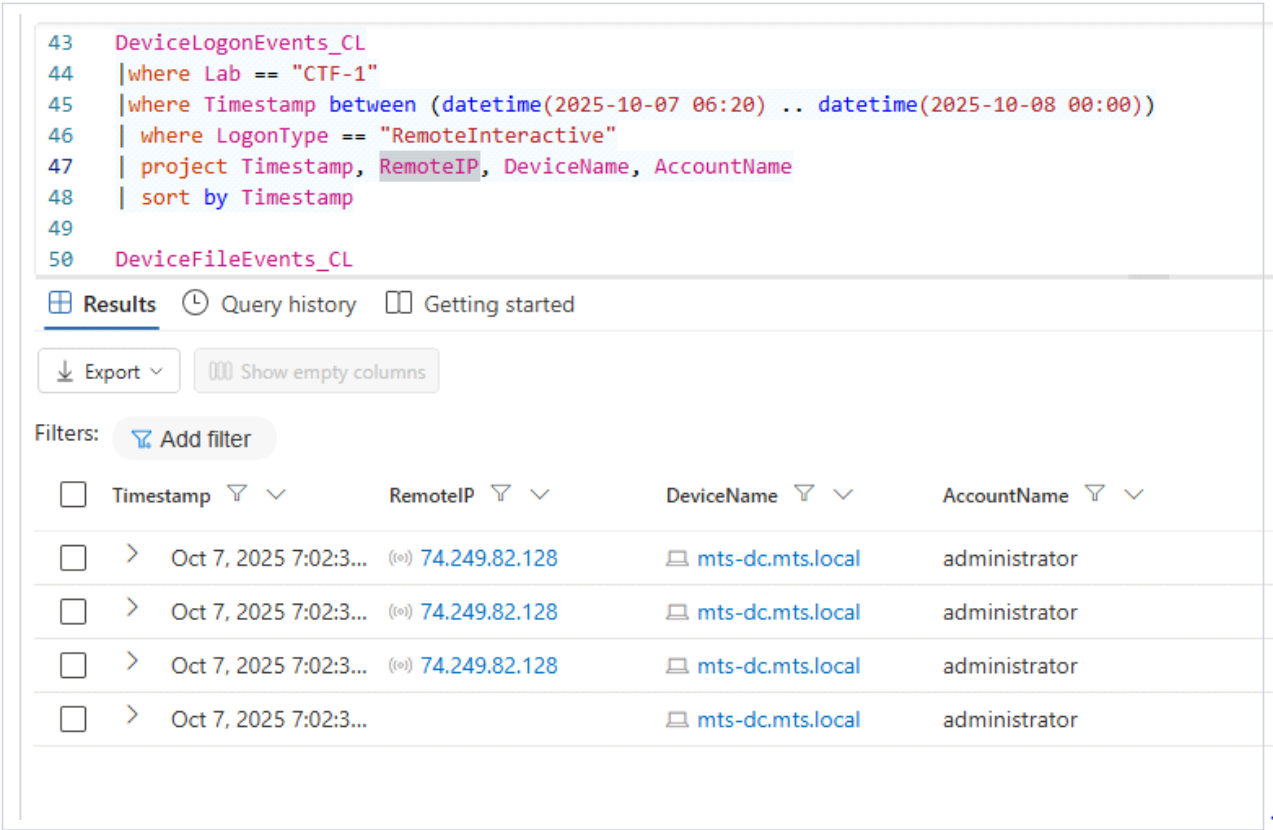


Figure E-10: Network and file telemetry showing the download and staging of mimikatz.exe in C:\Windows\Temp\TMP121235\ on MTS-CONTRACTORPC1.

E-11 — Domain Controller Authentication

☐	>	Oct 7, 2025 7:02:23 AM	"cmdkey.exe" /generic:20.246.83.211 /user:MTS\administrator /pass:*****
☐	>	Oct 7, 2025 7:02:23 AM	"cmdkey.exe" /delete:LegacyGeneric:target=Temp-MTS-DC
☐	>	Oct 7, 2025 7:02:24 AM	"mstsc.exe" /v:20.246.83.211



Figures E-11: Authentication telemetry showing the Administrator account accessing MTS-DC.mts.local after the compromise of MTS-CONTRACTORPC1.

E-12 — Active Directory Module Imported

Timestamp	ProcessCommandLine	FileName	InitiatingProcessFileName	InitiatingProcessCommandLine	FolderPath
Oct 7, 2025 7:04:2...	"powershell.exe" -noexit -command import-module ActiveDirectory	powershell.exe	explorer.exe	Explorer.EXE	C:\Windows\System32\WindowsPowerShell
Oct 7, 2025 7:04:3...	"devicecensus.exe" UserCxt	DeviceCensus.exe	svchost.exe	svchost.exe -k netsvcs -p -s Schedule	C:\Windows\System32\DeviceCensus.exe
Oct 7, 2025 7:04:4...	"MpCmdRun.exe" GetDeviceTicket -AccessKey 48F48E2E-73A5-F377-52A1-2BC5D5CA6B43	MpCmdRun.exe	mssmper.exe	MpEng.exe	C:\ProgramData\Microsoft\Windows Defen
Oct 7, 2025 7:05:5...	"MpCmdRun.exe" GetDeviceTicket -AccessKey 12F2C0FA-3719-3661-A7C4-F1F218F1F128	MpCmdRun.exe	mssmper.exe	MpEng.exe	C:\ProgramData\Microsoft\Windows Defen
Oct 7, 2025 7:07:0...	svchost_update.exe	svchost_update.exe	powershell.exe	"powershell.exe" -noexit -command import-module ActiveDirectory	C:\Users\Public\svchost_update.exe
Oct 7, 2025 7:07:0...	MicrosoftEdgeUpdate.exe	MicrosoftEdgeUpdate.exe	powershell.exe	"powershell.exe" -noexit -command import-module ActiveDirectory	C:\Users\Public\MicrosoftEdgeUpdate.exe
Oct 7, 2025 7:07:0...	quickbooks_sync.exe	quickbooks_sync.exe	powershell.exe	"powershell.exe" -noexit -command import-module ActiveDirectory	C:\Users\Public\quickbooks_sync.exe
Oct 7, 2025 7:07:2...	powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive -Command "& (\$Output...	powershell.exe	senseir.exe	"SenseIR.exe" "OfflineSenseIR" "1284" "eyJDb21hYXN5bW5kSWQhLCJCb...	C:\Windows\System32\WindowsPowerShell
Oct 7, 2025 7:07:2...	conhost.exe 0xffffffff -ForceV1	conhost.exe	powershell.exe	powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive ...	C:\Windows\System32\conhost.exe
Oct 7, 2025 7:07:4...	"netsh.exe" advfirewall set allprofiles state off	netsh.exe	powershell.exe	"powershell.exe" -noexit -command import-module ActiveDirectory	C:\Windows\System32\netsh.exe

Figure E-12: PowerShell process telemetry showing the execution of Import-Module ActiveDirectory on the MTS domain controller.

E-13 — Suspicious Executables on the Domain Controller

Timestamp	ProcessCommandLine	FileName	InitiatingProcessFileName	InitiatingProcessCommandLine	FolderPath
Oct 7, 2025 7:04:2...	"powershell.exe" -noexit -command import-module ActiveDirectory	powershell.exe	explorer.exe	Explorer.EXE	C:\Windows\System32\WindowsPowerSh...
Oct 7, 2025 7:04:3...	"devicecensus.exe" UserCtx	DeviceCensus.exe	svchost.exe	svchost.exe -k netsvc -p -s Schedule	C:\Windows\System32\DeviceCensus.exe
Oct 7, 2025 7:04:4...	"MpCmdRun.exe" GetDeviceTicket -AccessKey 4BF48E2E-73A5-F377-52A1-28C5DCA6B843	MpCmdRun.exe	mispeng.exe	MsMpEng.exe	C:\ProgramData\Microsoft\Windows Defen...
Oct 7, 2025 7:05:5...	"MpCmdRun.exe" GetDeviceTicket -AccessKey 12F2C0FA-3719-3661-A7C4-F1F218F1F128	MpCmdRun.exe	mispeng.exe	MsMpEng.exe	C:\ProgramData\Microsoft\Windows Defen...
Oct 7, 2025 7:07:0...	svchost.update.exe	svchost.update.exe	powershell.exe	"powershell.exe" -noexit -command import-module ActiveDirectory	C:\Users\Public\svchost_update.exe
Oct 7, 2025 7:07:0...	MicrosoftEdgeUpdate.exe	MicrosoftEdgeUpdate.exe	powershell.exe	"powershell.exe" -noexit -command import-module ActiveDirectory	C:\Users\Public\MicrosoftEdgeUpdate.exe
Oct 7, 2025 7:07:0...	quickbooks_sync.exe	quickbooks_sync.exe	powershell.exe	"powershell.exe" -noexit -command import-module ActiveDirectory	C:\Users\Public\quickbooks_sync.exe
Oct 7, 2025 7:07:2...	powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive -Command "& {Output...	powershell.exe	sensir.exe	"SenseIR.exe" -OfflineSensorID "1284" -ey/Db21YW5k5WQjOILCEb...	C:\Windows\System32\WindowsPowerShel...
Oct 7, 2025 7:07:2...	conhost.exe 0xfffffff -ForceV1	conhost.exe	powershell.exe	powershell.exe -ExecutionPolicy AllSigned -NoProfile -Noninteractive ...	C:\Windows\System32\conhost.exe
Oct 7, 2025 7:07:4...	"neth.exe" adfrewall set allprofiles state off	neth.exe	powershell.exe	"powershell.exe" -noexit -command import-module ActiveDirectory	C:\Windows\System32\neth.exe

Figure E-13: Process telemetry showing the execution of suspicious files on the domain controller, including svchost_update.exe, MicrosoftEdgeUpdate.exe and quickbooks_sync.exe.

E-14 — Sensitive File Access

[illegible]

Figure E-14: File-access telemetry showing multiple sensitive files being opened or accessed on the domain controller between approximately 7:17:54 AM and 8:00:35 AM.

E-15 — Volume Shadow Copy Created

```
25
26 DeviceEvents_CL
27 | where Lab == "CTF-1"
28 | where Timestamp between (datetime(2025-10-07 04:13).. datetime(2025-10-08 00:00))
29 | where DeviceName contains "mts-dc"
30 | where AdditionalData contains "vssadmin"
31 | project Timestamp, AdditionalData
32 | sort by Timestamp asc |
```

Results

Query history

Getting started

Export

Show empty columns

Filters:

Add filter

	Timestamp	AdditionalData
☐	> Oct 7, 2025 8:05:43 AM	["Command":Type-Command "\vssadmin.exe create shadow /for=C:\"]
☐	> Oct 7, 2025 8:16:05 AM	["Command":Type-Command "\vssadmin delete shadows /all /quiet"]

Figure E-15: Process telemetry showing vssadmin.exe creating a Volume Shadow Copy of the domain controller's C: drive.

E-16 — Active Directory Database Extraction



Figure E-16: Process telemetry showing ntdsutil.exe being used to extract Active Directory database information from the domain controller.

E-17 — Collected Data Archived

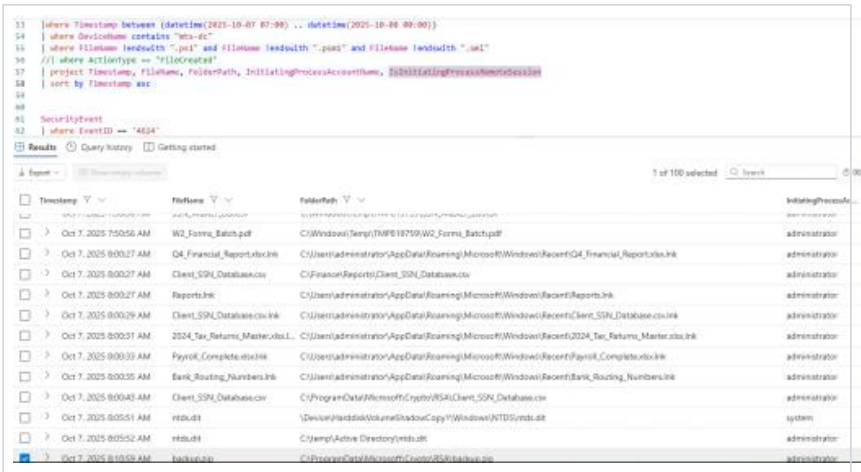


Figure E-17: File-creation telemetry showing collected information being archived into C:\ProgramData\Microsoft\Crypto\RSA\backup.zip.

E-18 — Possible Upload to File-Sharing Service

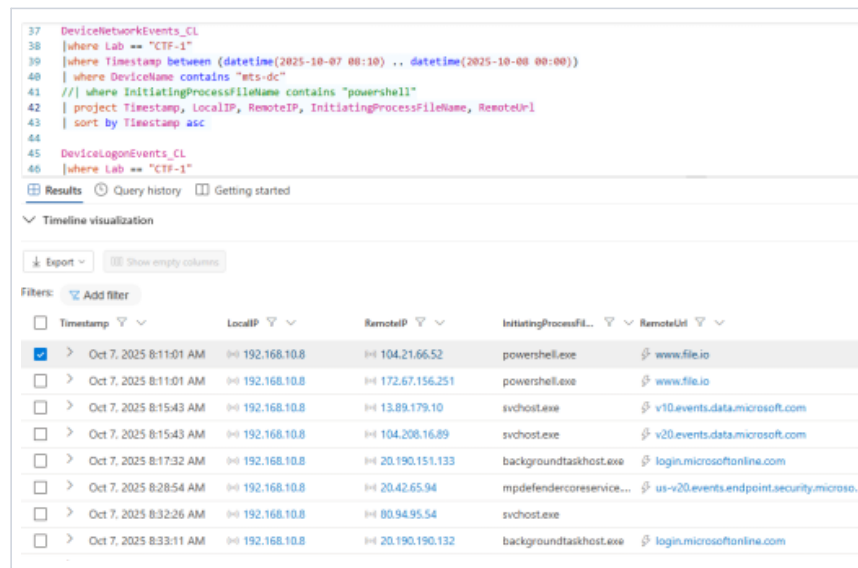


Figure E-18: Network telemetry showing communication with file.io shortly after the creation of backup.zip, indicating possible data exfiltration through a web-based file-sharing service.

E-19 — Possible DNS-Based Exfiltration



E-20 — Windows Event Logs Cleared

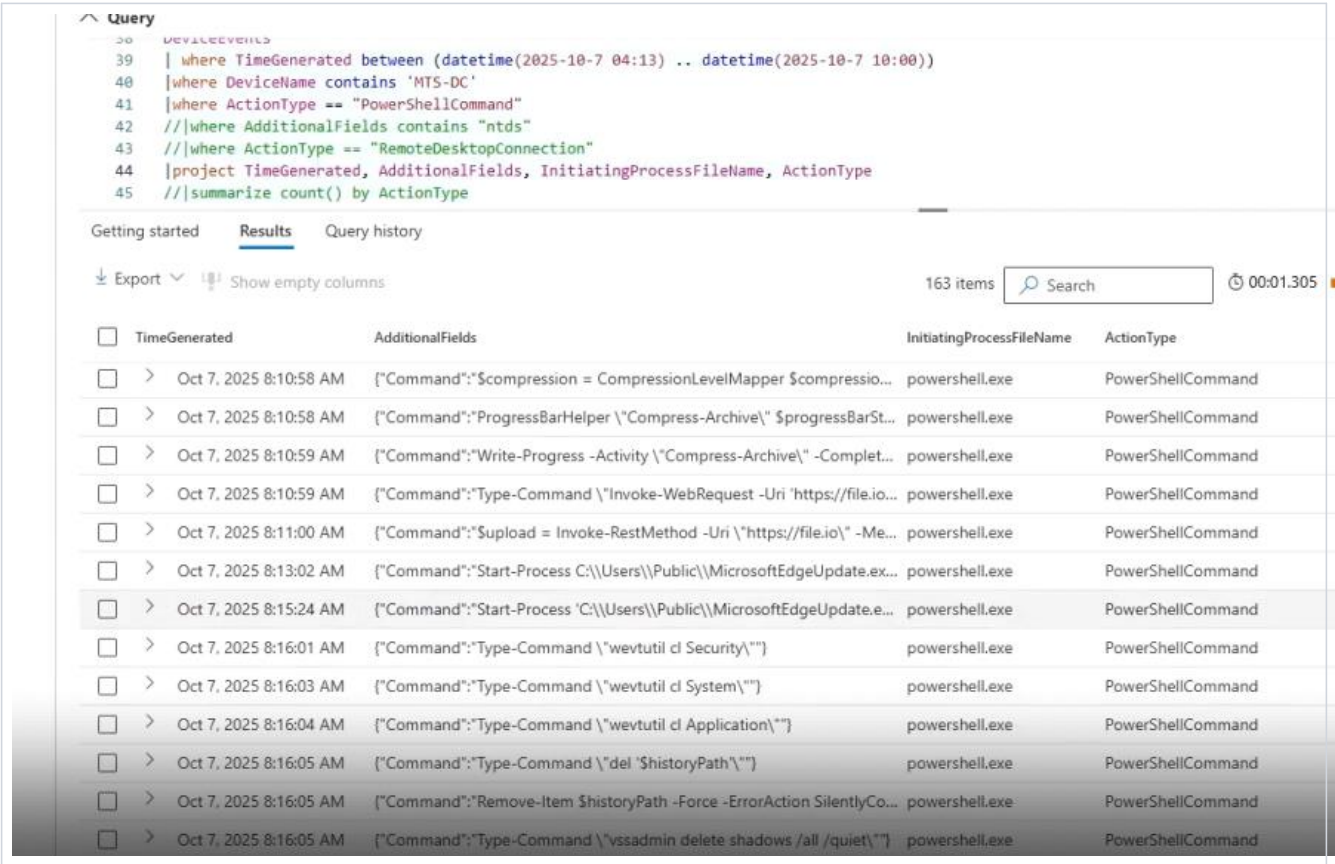


Figure E-20: Event or process telemetry showing the clearing of Windows Security, Application and System event logs on the affected system.

E-21 — Shadow Copies Deleted

Screenshot not included in the source report.

E-22 — PowerShell Command History Deleted

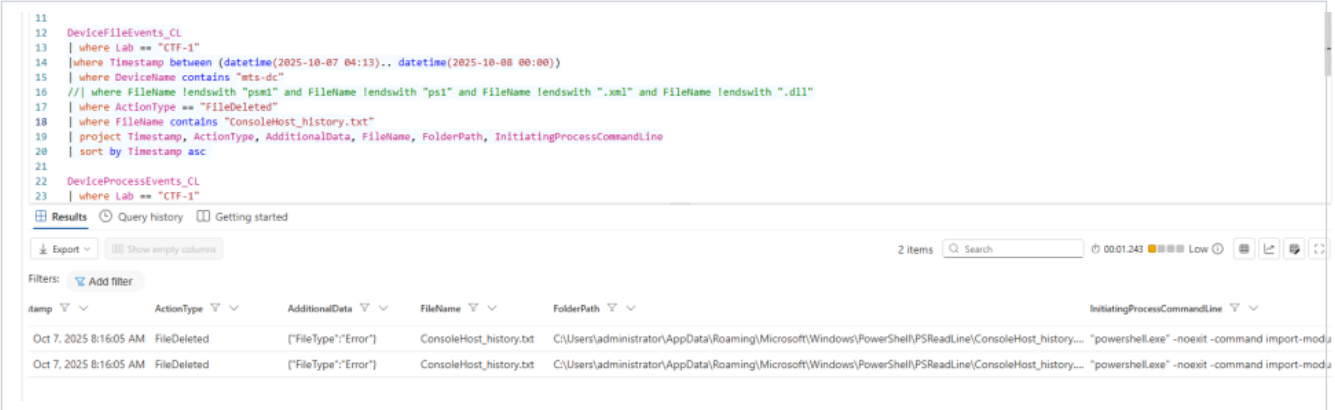


Figure E-22: File-deletion telemetry showing the removal of the PowerShell command-history file, consistent with an attempt to conceal previously executed commands.